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SUMMARY

IEEE-published Machine Learning Engineer and national hackathon winner (Industry Innovation '26) specializing in end-to-end AI systems, from fraud detection pipelines to RAG-based career platforms. Proficient in PyTorch, FastAPI, and LLM fine-tuning with production deployment experience via Docker and cloud-ready microservices. Seeking an entry-level ML/AI engineering role to build scalable, explainable AI systems.

EDUCATION

Sathyabama Institute of Science and Technology <i>M.Sc. Computer Science</i>	2025 – 2027 (In Progress)
Sathyabama Institute of Science and Technology <i>B.Sc. Computer Science (AI Specialization)</i>	2025

TECHNICAL SKILLS

Languages: Python, Java, SQL
Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Data Preprocessing, Cross Validation, Hyperparameter Tuning, Model Evaluation, Explainable AI (SHAP)
Deep Learning & NLP: PyTorch, TensorFlow, Neural Networks, CNNs, RNNs, LSTMs, Transformers, BERT, Sentence-Transformers
Generative AI & Retrieval: RAG, LangChain, Embeddings, Vector Databases, FAISS, ChromaDB, Semantic Search, Agentic AI, LLM Fine-Tuning (LLaMA)
Backend & APIs: FastAPI, REST APIs, Microservices
Data Engineering: EDA, Data Cleaning, Selenium, Pandas, NumPy
Computer Vision & 3D AI: OpenCV, Optical Flow, COLMAP, 3D Gaussian Splatting, FFmpeg
Databases: PostgreSQL, MongoDB
Cloud, DevOps & MLOps: AWS Fundamentals, Docker, Kubernetes, Git, GitHub, GitHub Actions, CI/CD, MLflow, Model Deployment
Frontend: React, Tailwind CSS, Streamlit

WORK EXPERIENCE

Production AI Developer (Intern) <i>InnoXR Labs Pvt. Ltd.</i>	Mar 2026 – Jun 2026 <i>Chennai, India</i>
- Developed an applied machine learning pipeline for single-video 3D reconstruction (MonoSplat) using COLMAP for camera pose estimation and Gaussian Splatting in PyTorch. - Worked on AI system design, building a hybrid CPU-GPU data preprocessing pipeline with FFmpeg for cloud-based execution. - Deployed a scalable FastAPI backend with async job handling and a Three.js viewer for end-to-end 3D visualization, within an immersive technology startup environment.	
Data Analyst Intern <i>India Advocacy</i>	Jul 2024 – Oct 2024 <i>Chennai, India</i>
Built Java and Selenium ETL pipelines to scrape and process 1M+ records; performed data preprocessing and cleaning to transform raw data into 15–20k structured datasets through multi-stage validation.	
Security Trooper (National Service) <i>Singapore Armed Forces</i>	Jan 2020 – Jan 2022 <i>Singapore</i>
- Executed security and vehicular operations in regulated environments, ensuring strict protocol compliance and operational readiness. - Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.	
Sales Associate (Part-Time) <i>Bata</i>	Feb 2022 – Aug 2022 <i>Singapore</i>
- Handled high-volume customer interactions in a fast-paced retail environment, ensuring efficient service delivery. - Managed billing, inventory coordination, and issue resolution to support smooth store operations.	

PROJECTS

FinGuard Pro: Real-Time Fraud Detection & Compliance System

IEEE ICAAIC 2025

- Developed a real-time UPI fraud detection pipeline trained on **100,000 synthetic transactions** using **25+ engineered features**, including geo-deviation scores and amount-to-average behavioral ratios.
- Evaluated **XGBoost**, **LightGBM**, and **CatBoost** using **Optuna TPE hyperparameter tuning**; selected **XGBoost** achieving **94% recall**, **ROC-AUC 0.9951**, and **F1-score 0.57**.
- Integrated **SHAP TreeExplainer** for local/global explainability and built a **fuzzy AML screening engine** using **RapidFuzz** over a **5,000-entity watchlist** with rule-based detection for structuring, round-tripping, and velocity anomalies.
- Built a dual-dashboard architecture using **FastAPI** and **Streamlit** with real-time alerts, SHAP visualizations, and automated PDF/ZIP compliance report generation, achieving **sub-3s inference latency**.

CareerGenie: AI Career Platform

Hackathon Winner, Industry Innovation '26

- Built an end-to-end **RAG-based career platform** using **MiniLM embeddings** and **ChromaDB** for semantic job matching beyond keyword-based retrieval.
- Designed a **multi-agent orchestration system** with specialized Resume, Job, Interview, Roadmap, and Advisor agents, including an **Agent Debate pipeline** (proposer–critic–synthesizer) to improve response reliability and reduce hallucinations.
- Implemented a **Learning-to-Rank recommendation engine** using **Bayesian Personalized Ranking (BPR)** trained on click, application, and offer signals with weighted semantic and skill-based ranking.
- Integrated a multi-provider LLM fallback pipeline (**Groq** → **Anthropic**) to improve response stability and reduce degraded-provider latency during inference.
- Delivered a full-stack deployment using **React** + **Tailwind**, **FastAPI**, **Netlify**, and **ngrok**, supporting ATS scoring, resume rewriting, mock interviews, and personalized learning roadmaps.

MonoSplat: Single-Video 3D Reconstruction Pipeline

Internship Project, 2026

- Engineered a monocular 3D reconstruction pipeline using **COLMAP**, **Gaussian Splatting**, and **PyTorch** for reconstructing interactive 3D scenes from single-video input.
- Designed a hybrid **CPU-GPU preprocessing pipeline** using **FFmpeg** to optimize frame extraction and cloud-based processing workflows.
- Developed an asynchronous **FastAPI** backend integrated with a **Three.js** visualization interface for interactive browser-based rendering.
- Optimized preprocessing and rendering workflows to improve usability and reduce end-to-end reconstruction overhead.

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection

Design Thinking Project, 2026

- Developed a privacy-preserving crowd anomaly detection system using **Farneback optical flow** and a **2×3 zone-grid architecture** to extract density, velocity, and directional variance metrics without tracking individuals.
- Conducted cross-model evaluation across **5 unsupervised anomaly detection models** (MAD, Z-Score, One-Class SVM, Isolation Forest, and LSTM Autoencoder) trained on **15,928 ETH/UCY records** and validated on **13,805 Zara1/Zara2 records**.
- Selected **Median Absolute Deviation (MAD)** as the primary detector after cross-domain testing on the Mall dataset, achieving **ROC-AUC 0.8391** and reducing false positives by **40–60%** through adaptive camera-shake filtering.
- Built a **FastAPI** backend and **Streamlit** dashboard with explainable real-time alerts and zone-level reasoning for crowd safety monitoring.

SourceUp: Explainable Supplier Recommendation System

Final Year Capstone, 2026

- Implemented semantic supplier retrieval using **Sentence-BERT** embeddings and **FAISS** vector search for natural-language supplier discovery.
- Designed a ranking pipeline using **Scikit-learn** incorporating cost, delivery timelines, and supplier reliability constraints.
- Integrated an explainability layer to provide feature-level reasoning and risk insights for supplier recommendation decisions.

PUBLICATIONS & CONFERENCES

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025*

ISBN: 979-8-3315-6587-9

ACHIEVEMENTS & HACKATHONS

Industry Innovation Hackathon <i>Sathyabama Institute of Science and Technology – National Level</i>	Mar 2026
• Won Best Industry Innovation Award at Industry Innovation '26.	
Origin 24-Hour Hackathon <i>SIMATS Engineering, Chennai</i>	Apr 2026
• Placed 4th at Origin 24-Hour Hackathon.	

CERTIFICATIONS

AI & ML <i>Space Zee Technologies</i>	Jul 2024
Python Core <i>Space Zee Technologies</i>	Feb 2024